

Not In Our Time: Transport-Based Content Selection as an Inspectable, Manipulable Alternative to RAG

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We present an alternative to Retrieval-Augmented Generation [Lewis et al., 2020] in which passage selection is a min-cost flow problem [Ahuja et al., 1993, Peyré and Cuturi, 2019]. Min-cost flow is cheap compared to LLM regeneration, so the system gives a podcast producer a flexible thinking tool for rapid exploration of alternatives. We demonstrate it by generating imitation panel discussions in the style of BBC Radio 4’s *In Our Time* [Bragg, 1998–present], about Dickens’ *Bleak House* [Dickens, 1852–1853]. RAG as typically deployed [Gao et al., 2024]—including in Google’s NotebookLM podcast generator [Google, 2024]—retrieves by embedding similarity and feeds results to an LLM. This is effective but opaque: the user cannot see *why* a passage was selected, cannot adjust selection criteria, and cannot explore how different editorial priorities reshape the output.

The system operates over 6,916 passages enriched by an LLM with structured metadata and contextual embeddings [Anthropic, 2024] (20 fields covering narrative function, thematic weight, and emotional register). The first transport problem selects passages to satisfy *demand profiles* declared by fictional expert personas—a novelist, a legal historian, a close reader, a performer, a Marxist, a traditionalist. The second assigns them to episode segments. An LLM generates multi-voice dialogue with speaking-style annotations for TTS rendering.

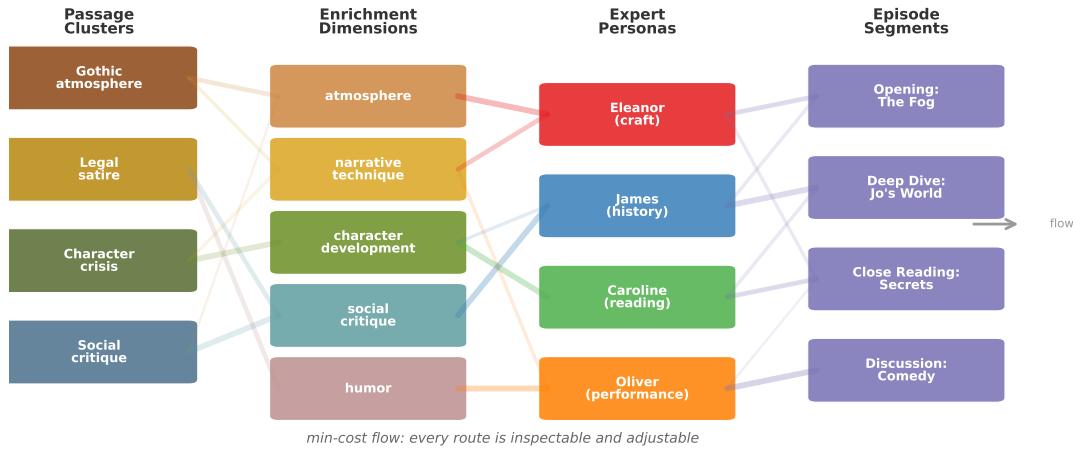


Figure 1: Passages flow through enrichment dimensions and expert demand nodes to episode segments. Every arc is inspectable and adjustable.

The expert personas are, deliberately, *stereotypes*: the Marxist always finds class struggle; the performer always finds comedy; the close reader always finds prose rhythm. The generated discussions are closer to parody than scholarship—each configuration produces a recognisably different *caricature* of academic discourse. This is the point. By making the stereotypes explicit and adjustable—encoded as demand vectors rather than implicit in a prompt—the system turns editorial bias into a first-class object that can be inspected, compared, and argued about. Every editorial decision is inspectable and manipulable. Twenty configurations, varying panel composition and arc emphasis, produce measurably different outputs: different passage selections (mean pairwise Jaccard similarity 0.52), different vocabulary, and different grammatical registers, confirmed by corpus comparison methods [Kilgariff, 2001]. What happens when we remove the social historian? Double a character arc? Replace the close reader with a comedian? The answers are computable, and the differences are legible.

We do not claim the generated podcasts are good—they are fluent, structured, and plausible-sounding, which is precisely the problem with LLM-generated academic discussion [Doshi and Hauser, 2024]. Following Boden’s [2004] taxonomy, the system performs combinational creativity, but the *interesting* creativity is the user’s: exploring the configuration space and confronting the stereotypes that make the machinery work. The system’s value is diagnostic, turning a black-box pipeline into a navigable space of configurations. It is no accident that *Bleak House* centres on the law: lawyers reviewing documents under constraints need inspectable, manipulable thinking tools at least as much as literary critics do.

Keywords: optimal transport, min-cost flow, RAG, LLM stereotypes, editorial decision spaces

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